# CLASS 7 — MATHS · FORMULA SHEET

Chapter 3: **A Peek Beyond the Point** | Quick revision, one page

## Place value of decimals

- Places after the point: tenths $\left(\frac{1}{10}\right)$, hundredths $\left(\frac{1}{100}\right)$, thousandths $\left(\frac{1}{1000}\right)$.
- $0.375 = \frac{3}{10} + \frac{7}{100} + \frac{5}{1000} = \frac{375}{1000}$. Zeros at the end of the decimal part do not change the value: $2.50 = 2.5$.

## Comparing and rounding

- Compare whole parts first, then tenths, hundredths, ... digit by digit (give both numbers equal decimal places).
- Rounding: look at the digit just after the required place; $5$ or more rounds up, less than $5$ rounds down.

## Addition and subtraction

- Write the numbers with the decimal points in a line and add or subtract as whole numbers.

## Multiplication

- Multiply ignoring the points; decimal places in the product $=$ sum of decimal places in the factors.
- Multiplying by $10, 100, 1000$ moves the point $1, 2, 3$ places to the right.

## Division

- Dividing by a decimal: multiply dividend and divisor by $10, 100, \ldots$ to make the divisor a whole number, e.g. $4.5 \div 0.5 = 45 \div 5$.
- Dividing by $10, 100, 1000$ moves the point $1, 2, 3$ places to the left.

## Units

- $1\ \text{m} = 100\ \text{cm}$, $1\ \text{cm} = 0.01\ \text{m}$; $1\ \text{kg} = 1000\ \text{g}$; $1\ \text{L} = 1000\ \text{mL}$; ₹$1 = 100$ paise.

MASTERMIND